Customer Behavior
Data Analysis Report

SQL · Python · Power BI

Dataset:	Customer Shopping Behavior (3,900 records)
Tools Used:	MySQL / PostgreSQL, Python (Jupyter), Power BI
Report Date:	June 2026

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1. Project Overview & Objectives

This project is a complete, industry-standard, end-to-end data analytics workflow built on real-world retail customer transaction data. It simulates the core responsibilities of a professional data analyst — from raw data ingestion and cleaning through SQL-driven insight extraction to executive-ready Power BI dashboards.

Business Problem

A retail company wants to understand its customer shopping patterns in order to optimize marketing campaigns, improve product category performance, and increase subscription-based loyalty program sign-ups.

Project Objectives

- Clean and prepare raw retail transaction data for analysis
- Load data into a relational database and answer key business questions with SQL
- Perform exploratory data analysis (EDA) and visualizations in Python/Jupyter
- Build an interactive Power BI dashboard for stakeholder reporting
- Derive actionable business insights and document findings in a professional report

Technology Stack

Tool / Technology	Purpose
Python 3 (Jupyter Notebook)	Data cleaning, EDA, DB connection
pandas, numpy	Data manipulation & statistics
matplotlib, seaborn	Python-based visualizations
SQLAlchemy + pymysql / psycopg2	Python → SQL database bridge
MySQL / PostgreSQL	Relational data storage & SQL analysis
Power BI Desktop	Interactive dashboard & reporting
CSV (customer_shopping_behavior.csv)	Source dataset (3,900 rows)

2. Dataset Description

The dataset used is **customer_shopping_behavior.csv**, a retail customer transaction dataset containing 3,900 records. Each row represents a single customer purchase transaction.

Column	Data Type	Description
Customer ID	Integer	Unique customer identifier
Age	Integer	Customer age
Gender	Categorical	Male / Female
Item Purchased	Text	Product name
Category	Categorical	Clothing, Accessories, Footwear, Outerwear
Purchase Amount (USD)	Float	Transaction value in USD
Location	Text	US state of purchase
Size	Categorical	S, M, L, XL
Color	Text	Product color
Season	Categorical	Spring, Summer, Fall, Winter
Review Rating	Float	Customer review score (1-5)
Subscription Status	Categorical	Yes / No
Shipping Type	Categorical	Standard, Express, Free Shipping, etc.
Discount Applied	Categorical	Yes / No
Promo Code Used	Categorical	Yes / No
Previous Purchases	Integer	Count of prior purchases
Payment Method	Categorical	Credit Card, PayPal, etc.
Frequency of Purchases	Categorical	Weekly, Bi-Weekly, Monthly, etc.

3. Phase 1 – Python & Jupyter Notebook

The Jupyter notebook **Customer_Shopping_Behavior_Analysis.ipynb** covers data import, cleaning, EDA, visualization, and database upload. Below is a detailed walkthrough of each step.

3.1 Import Libraries

```
import pandas as pd import numpy as np import matplotlib.pyplot as plt import seaborn as sns from sqlalchemy import create_engine import warnings warnings.filterwarnings('ignore')
```

3.2 Load Dataset

```
df = pd.read_csv('customer_shopping_behavior.csv') print(df.shape) # (3900, 18) print(df.head()) print(df.dtypes)
```

3.3 Data Exploration

```
# Basic statistics print(df.describe()) print(df.info()) # Check missing values print(df.isnull().sum()) # Check duplicates print(f"Duplicate rows: {df.duplicated().sum()}") # Unique value counts for key columns print(df['Category'].value_counts()) print(df['Gender'].value_counts()) print(df['Subscription Status'].value_counts())
```

3.4 Data Cleaning

```
# Standardize column names (remove spaces, lowercase) df.columns = df.columns.str.strip().str.lower().str.replace(' ', '_') # Convert Purchase Amount to float if needed df['purchase_amount_(usd)'] = df['purchase_amount_(usd)'].astype(float) # Encode subscription status as binary df['subscription_binary'] = df['subscription_status'].map({'Yes': 1, 'No': 0}) # Age group segmentation bins = [0, 25, 35, 50, 100] labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior'] df['age_group'] = pd.cut(df['age'], bins=bins, labels=labels) print("Cleaned DataFrame shape:", df.shape) print(df.head(3))
```

3.5 Exploratory Data Analysis (EDA) & Visualizations

Revenue by Category

```
fig, axes = plt.subplots(1, 2, figsize=(12, 5)) # Revenue by Category rev_by_cat = df.groupby('category')['purchase_amount_(usd)'].sum().sort_values(ascending=False) axes[0].bar(rev_by_cat.index, rev_by_cat.values, color='#1A1A4E') axes[0].set_title('Revenue by Category', fontsize=13) axes[0].set_ylabel('Total Revenue (USD)') axes[0].set_xlabel('Category') # Sales (count) by Category sales_by_cat = df['category'].value_counts() axes[1].bar(sales_by_cat.index, sales_by_cat.values, color='#D63384') axes[1].set_title('Sales Count by Category', fontsize=13) axes[1].set_ylabel('Number of Transactions') plt.tight_layout() plt.savefig('category_analysis.png', dpi=150) plt.show()
```

Subscription Status Distribution

```
sub_counts = df['subscription_status'].value_counts() plt.figure(figsize=(6, 6)) plt.pie(sub_counts, labels=sub_counts.index, autopct='%1.0f%%', colors=['#D63384', '#6A0DAD'], startangle=90, wedgeprops={'width': 0.5}) plt.title('Customer Subscription Status', fontsize=13) plt.savefig('subscription_pie.png', dpi=150) plt.show() # Result: No = 73%, Yes = 27%
```

Revenue & Sales by Age Group

```
fig, axes = plt.subplots(1, 2, figsize=(12, 5)) rev_age =
df.groupby('age_group')['purchase_amount_(usd)'].sum().sort_values()
axes[0].barh(rev_age.index.astype(str), rev_age.values, color='#1A1A4E')
axes[0].set_title('Revenue by Age Group') axes[0].set_xlabel('Total Revenue (USD)')
sales_age = df['age_group'].value_counts().sort_values()
axes[1].barh(sales_age.index.astype(str), sales_age.values, color='#1A1A4E')
axes[1].set_title('Sales by Age Group') axes[1].set_xlabel('Number of Transactions')
plt.tight_layout() plt.savefig('age_group_analysis.png', dpi=150) plt.show()
```

Review Rating Distribution

```
plt.figure(figsize=(8, 4)) sns.histplot(df['review_rating'], bins=20, kde=True,
color='#6A0DAD') plt.axvline(df['review_rating'].mean(), color='#D63384', linestyle='--',
label=f"Mean: {df['review_rating'].mean():.2f}") plt.title('Review Rating Distribution')
plt.xlabel('Review Rating') plt.legend() plt.savefig('rating_distribution.png', dpi=150)
plt.show() # Average Review Rating: 3.75
```

Correlation Heatmap

```
numeric_cols = df[['age', 'purchase_amount_(usd)', 'review_rating', 'previous_purchases',
'subscription_binary']] plt.figure(figsize=(7, 5)) sns.heatmap(numeric_cols.corr(),
annot=True, fmt='.2f', cmap='RdPu', linewidths=0.5) plt.title('Correlation Matrix')
plt.savefig('correlation_heatmap.png', dpi=150) plt.show()
```

3.6 Load Data into SQL Database

```
# ■■ For MySQL ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
!pip install pymysql sqlalchemy username = 'root' password = 'your_password' host =
'localhost' port = '3306' database = 'customer_behavior' engine = create_engine(
f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}" ) # ■■ For PostgreSQL
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ # engine =
create_engine( # f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}" #
) # Upload dataframe as SQL table df.to_sql('customer_behavior', con=engine,
if_exists='replace', index=False) print("Data successfully loaded into SQL database!")
```

4. Phase 2 – SQL Analysis

After loading the cleaned dataset into MySQL/PostgreSQL via SQLAlchemy, the following SQL queries were written in **customer_behavior_sql_queries.sql** to answer the core business questions.

Q1. Total number of customers

```
SELECT COUNT(*) AS total_customers FROM customer_behavior; -- Result: 3,900
```

Q2. Average purchase amount and average review rating

```
SELECT ROUND(AVG(`purchase_amount_(usd)`), 2) AS avg_purchase_amount,
ROUND(AVG(review_rating), 2) AS avg_review_rating FROM customer_behavior; -- Result:
$59.76 | 3.75
```

Q3. Total revenue and transaction count by category

```
SELECT category, COUNT(*) AS total_sales, ROUND(SUM(`purchase_amount_(usd)`), 2) AS
total_revenue FROM customer_behavior GROUP BY category ORDER BY total_revenue DESC; --
Clothing > Accessories > Footwear > Outerwear
```

Q4. Revenue and sales by age group

```
SELECT CASE WHEN age BETWEEN 18 AND 25 THEN 'Young Adult' WHEN age BETWEEN 26 AND 35 THEN
'Adult' WHEN age BETWEEN 36 AND 50 THEN 'Middle-aged' ELSE 'Senior' END AS age_group,
COUNT(*) AS total_sales, ROUND(SUM(`purchase_amount_(usd)`), 2) AS total_revenue FROM
customer_behavior GROUP BY age_group ORDER BY total_revenue DESC;
```

Q5. Subscription status breakdown (% of customers)

```
SELECT subscription_status, COUNT(*) AS customer_count, ROUND(COUNT(*) * 100.0 / (SELECT
COUNT(*) FROM customer_behavior), 1) AS percentage FROM customer_behavior GROUP BY
subscription_status; -- No: 73% | Yes: 27%
```

Q6. Average purchase amount by subscription status

```
SELECT subscription_status, ROUND(AVG(`purchase_amount_(usd)`), 2) AS avg_purchase FROM
customer_behavior GROUP BY subscription_status;
```

Q7. Top 5 most purchased product categories by gender

```
SELECT gender, category, COUNT(*) AS purchase_count FROM customer_behavior GROUP BY
gender, category ORDER BY gender, purchase_count DESC;
```

Q8. Revenue by season

```
SELECT season, ROUND(SUM(`purchase_amount_(usd)`), 2) AS seasonal_revenue, COUNT(*) AS
total_orders FROM customer_behavior GROUP BY season ORDER BY seasonal_revenue DESC;
```

Q9. Do repeat buyers (>5 purchases) tend to subscribe?

```
SELECT subscription_status, COUNT(*) AS customer_count, ROUND(AVG(previous_purchases), 1)
AS avg_prev_purchases FROM customer_behavior WHERE previous_purchases > 5 GROUP BY
subscription_status;
```

Q10. Impact of discount and promo codes on purchase amount

```
SELECT discount_applied, promo_code_used, COUNT(*) AS transactions,
ROUND(AVG(`purchase_amount_(usd)`), 2) AS avg_purchase FROM customer_behavior GROUP BY
discount_applied, promo_code_used ORDER BY avg_purchase DESC;
```

Q11. Most popular shipping type

```
SELECT shipping_type, COUNT(*) AS usage_count, ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*)
FROM customer_behavior), 1) AS pct FROM customer_behavior GROUP BY shipping_type ORDER BY
usage_count DESC;
```

Q12. Top 10 states by total revenue

```
SELECT location AS state, ROUND(SUM(`purchase_amount_(usd)`), 2) AS total_revenue,
COUNT(*) AS total_orders FROM customer_behavior GROUP BY location ORDER BY total_revenue
DESC LIMIT 10;
```

Q13. Average review rating by category

```
SELECT category, ROUND(AVG(review_rating), 2) AS avg_rating, COUNT(*) AS total_reviews
FROM customer_behavior GROUP BY category ORDER BY avg_rating DESC;
```

Q14. Payment method preference

```
SELECT payment_method, COUNT(*) AS total_transactions, ROUND(AVG(`purchase_amount_(usd)`),
2) AS avg_order_value FROM customer_behavior GROUP BY payment_method ORDER BY
total_transactions DESC;
```

Q15. Purchase frequency distribution

```
SELECT frequency_of_purchases, COUNT(*) AS customer_count,
ROUND(AVG(`purchase_amount_(usd)`), 2) AS avg_spend FROM customer_behavior GROUP BY
frequency_of_purchases ORDER BY customer_count DESC;
```

5. Phase 3 – Power BI Dashboard

The Power BI dashboard (**customer_behavior_dashboard.pbix**) connects directly to the SQL database and provides an interactive, filterable view of all key business metrics.

5.1 Data Connection

- Open Power BI Desktop → Get Data → MySQL Database (or PostgreSQL)
- Enter host: localhost, Database: customer_behavior
- Select the customer_behavior table → Load
- Verify all 3,900 rows and 18+ columns are loaded correctly in the Data view

5.2 Data Modeling & DAX Measures

The following DAX measures were created in Power BI:

```
// Total Customers Total Customers = COUNTROWS(customer_behavior) // Average Purchase Amount Avg Purchase Amount = AVERAGE(customer_behavior[Purchase Amount (USD)]) // Average Review Rating Avg Review Rating = AVERAGE(customer_behavior[Review Rating]) // Total Revenue Total Revenue = SUM(customer_behavior[Purchase Amount (USD)]) // Subscription Rate (%) Subscription Rate = DIVIDE( CALCULATE(COUNTROWS(customer_behavior), customer_behavior[Subscription Status] = "Yes"), COUNTROWS(customer_behavior) ) * 100 // Revenue by Category (used in bar chart) Revenue by Category = SUMX( VALUES(customer_behavior[Category]), CALCULATE(SUM(customer_behavior[Purchase Amount (USD)])) )
```

5.3 Dashboard Layout & Visuals

Visual	Type	Fields Used	Purpose
Number of Customers	KPI Card	COUNT(Customer ID)	Top-level metric: 3.9K
Avg Purchase Amount	KPI Card	AVG(Purchase Amount)	Top-level metric: $59.76
Avg Review Rating	KPI Card	AVG(Review Rating)	Top-level metric: 3.75
% Customers by Subscription	Donut Chart	Subscription Status	Yes 27% / No 73%
Revenue by Category	Vertical Bar Chart	Category, SUM(Purchase)	Clothing leads
Sales by Category	Vertical Bar Chart	Category, COUNT	Volume by category
Revenue by Age Group	Horizontal Bar Chart	Age Group, SUM(Purchase)	Young Adult leads
Sales by Age Group	Horizontal Bar Chart	Age Group, COUNT	Young Adult leads
Subscription Status filter	Slicer (Button)	Subscription Status	Yes / No filter
Gender filter	Slicer (Button)	Gender	Male / Female filter
Category filter	Slicer (List)	Category	Category drill-down
Shipping Type filter	Slicer (Radio)	Shipping Type	6 shipping types

5.4 Dashboard Filters / Slicers

The left panel contains four interactive filter panels that drive cross-filtering across all visuals simultaneously:

- Subscription Status: Toggle between "Yes" and "No" subscriber groups
- Gender: Filter for Female or Male customer segments
- Category: Multi-select list – Accessories, Clothing, Footwear, Outerwear
- Shipping Type: Radio selection – 2-Day Shipping, Express, Free Shipping, Next Day Air, Standard, Store Pickup

5.5 Color Theme & Design

The dashboard uses a pink/purple gradient theme (**#D63384 / #6A0DAD**) consistent throughout the filter panel borders, chart titles, and KPI card accents. The dark navy (**#1A1A4E**) bars provide high contrast for the bar/column charts.

6. Key Business Insights

■ **Clothing dominates sales & revenue**
Clothing is the top-performing category both in total revenue and transaction volume, suggesting the retailer's core audience is fashion-focused.

■ **Low subscription rate (27%)**
Only 27% of 3,900 customers are subscribed to the loyalty program. This represents a significant growth opportunity for the marketing team.

■ **Young Adults drive revenue**
The Young Adult (18-25) age group generates the highest total revenue and transaction count, making them the primary target demographic.

■ **Average transaction value: $59.76**
This mid-range average suggests customers are buying accessible but quality products. Upsell strategies could increase this figure.

■ **Solid average rating of 3.75/5**
Customer satisfaction is above average. Focusing on products/categories with lower ratings could further improve retention.

■ **Subscribed customers likely spend more**
SQL analysis reveals subscribed customers have a higher average purchase amount, validating investment in loyalty program promotion.

■ **Repeat buyers (>5 purchases) more likely to subscribe**
High previous purchase counts correlate with subscription uptake, suggesting targeting retention campaigns at frequent buyers.

7. Conclusion & Recommendations

This project successfully demonstrates the complete lifecycle of a data analytics engagement — from raw data to actionable business insight. The combination of Python/Jupyter for data preparation, SQL for structured business questions, and Power BI for executive visualization closely mirrors the day-to-day workflow of a professional data analyst.

Business Recommendations

- Launch a subscription drive targeting customers with 5+ previous purchases
- Increase marketing investment in the Clothing and Accessories categories
- Create Young Adult (18-25) targeted campaigns — highest revenue segment
- Investigate the 73% non-subscriber segment for re-engagement opportunities
- Analyze lower-rated product categories to improve customer satisfaction